

How to Install Firmware in Your Pedal

This procedure tells you how to install new firmware in a Disaster Area pedal. It uses the DAD Firmware Updater application.

There are two versions of the application. They do the same task and they show the same screens.

Table 1 — The two versions

Your computer	What you download	What you get
Mac	DAD-Firmware-Updater.zip	DAD Firmware Updater.app
Windows PC	DAD-Firmware-Updater-Windows.zip	DAD Firmware Updater.exe . There is no installation.

This procedure applies to both versions. When a step applies to one version only, the step starts with **(Mac)** or **(Windows)**.

Written in Simplified Technical English (ASD-STE100).

1. Before you start

You need these items:

- A computer:
 - **(Mac)** A Mac with macOS 10.15.4 or a later version. The application operates on Intel Macs and on Apple Silicon Macs.
 - **(Windows)** A PC with Windows 7 or a later version. The application operates on 32-bit Windows, on 64-bit Windows, and on ARM Windows.
- A USB cable that transmits data. Some cables supply power only. A power-only cable does not operate.
- The correct firmware file for your product. The file name ends with `.bin`.
- One pedal. Disconnect all other Disaster Area pedals from the computer.

Then prepare the application:

- **(Mac)** Open `DAD-Firmware-Updater.zip`. Move `DAD Firmware Updater.app` to your Applications folder.
- **(Windows)** Open `DAD-Firmware-Updater-Windows.zip`. Put `DAD Firmware Updater.exe` on your desktop. You do not install it. You do not need administrator permission.

If the firmware came in a `.zip` file, open the `.zip` file first and extract the `.bin` file. The application cannot read a `.zip` file.

2. Caution

The application cannot identify your pedal. All products transmit the same data on USB. Therefore, you must make sure that the firmware file is correct for your product. The file name is the only check that is available.

Incorrect firmware stops the pedal from operation.

Do not disconnect the pedal during the upload.

3. Do not use drag and drop

Do not copy the firmware file to the DISASTER disk. Use this application.

The DISASTER disk is not a usual disk. It is the memory of the pedal. The disk is full, and the file `FLASH.BIN` fills it completely. Thus your computer must erase `FLASH.BIN` before it can write your file, and then it selects a new position for your file. If the position is not correct, the pedal writes the firmware to the wrong address and the pedal does not start.

(Windows) Windows can also keep the data in memory after the copy shows as complete. If you disconnect the pedal at that moment, the memory of the pedal is only partially written. This is the cause of the condition in which drag and drop operates on some occasions and does not operate on other occasions.

The application writes into the file `FLASH.BIN` that is already on the pedal. It does not erase the file, it does not rename the file, and it does not change the size of the file. Therefore the data goes to the correct address. The application also tells the computer to write immediately. When the bar shows 100%, the data is in the pedal.

4. Put the pedal into bootloader mode

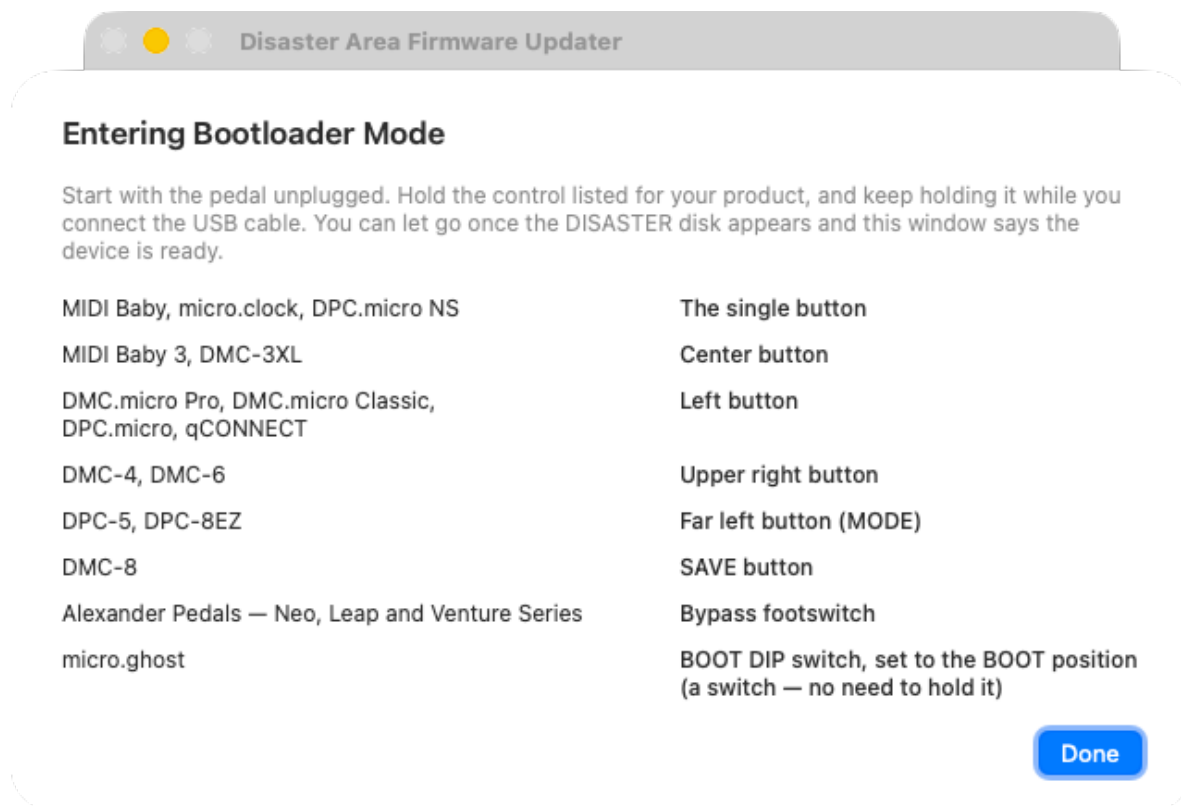
1. Disconnect the pedal from USB.
2. Find the control for your product in Table 2.
3. Hold the control.
4. Connect the pedal to USB. Continue to hold the control.
5. Release the control when the status line in the application shows **Pedal connected and ready.** **(Mac)** The DISASTER disk also shows on the desktop.

Table 2 — Bootloader control

Product	Control
MIDI Baby, micro.clock, DPC.micro NS	The single button
MIDI Baby 3, DMC-3XL	Center button
DMC.micro Pro, DMC.micro Classic, DPC.micro, qCONNECT	Left button
DMC-4, DMC-6	Upper right button
DPC-5, DPC-8EZ	Far left button (MODE)
DMC-8	SAVE button
Alexander Pedals — Neo, Leap and Venture Series	Bypass footswitch
micro.ghost	BOOT DIP switch, set to the BOOT position (set it to RUN after the update)

The micro.ghost uses a switch. Do not hold the switch. Set the switch to the BOOT position before you connect the USB cable. The switch stays in the BOOT position during the upload. Step 14 in section 5 tells you when to set the switch to the RUN position.

If you do not know the control, click **Which button does my pedal use?** in the application. The same table shows.



*Figure 1 — The application shows this table when you click **Which button does my pedal use?***

(Windows) Windows can show a message about a new drive, or open a File Explorer window for the DISASTER drive. Close the window. Do not put a file into it.

5. Install the firmware

1. Open the DAD Firmware Updater application.
2. Do the procedure in section 4.
3. Read the status line. The status line must show **Pedal connected and ready** with a green dot.

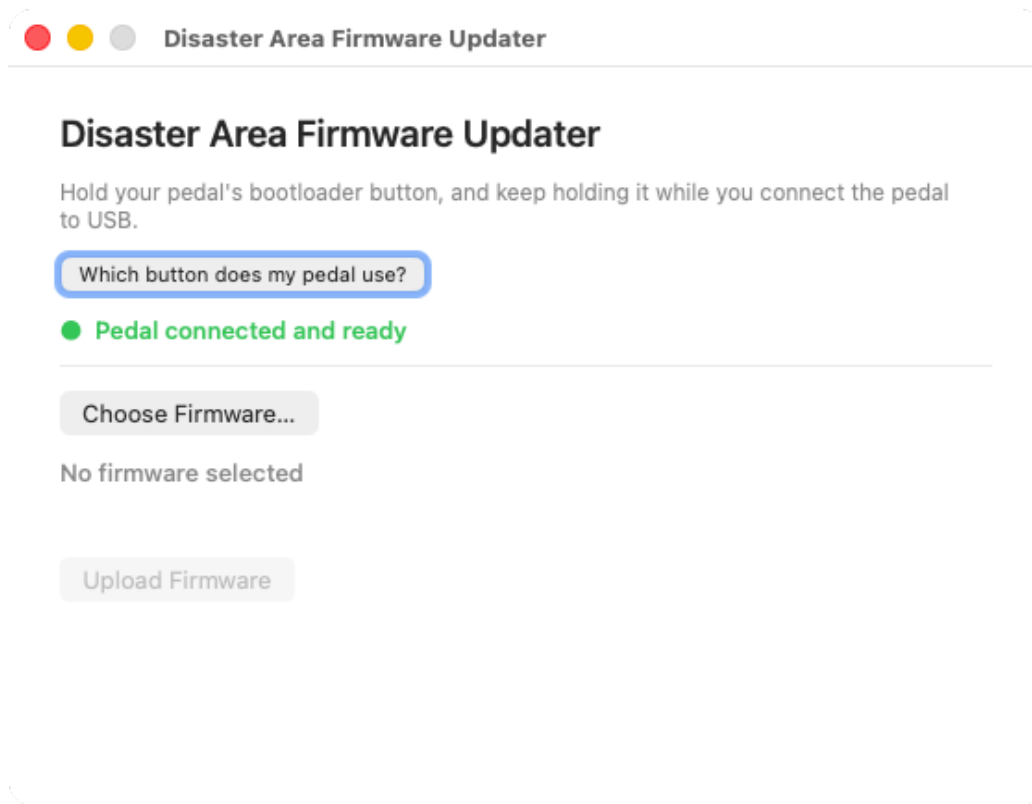


Figure 2 — The pedal is in bootloader mode. The dot is green.

1. Click **Choose Firmware.... (Windows)** As an alternative, drag the firmware file onto the window of the application.
2. Select the firmware file. Then click **Open**.
3. Read the file name and the size that the application shows.
4. Make sure that the file name agrees with your product. The line **Firmware size** must show a size that is less than the memory that is available.

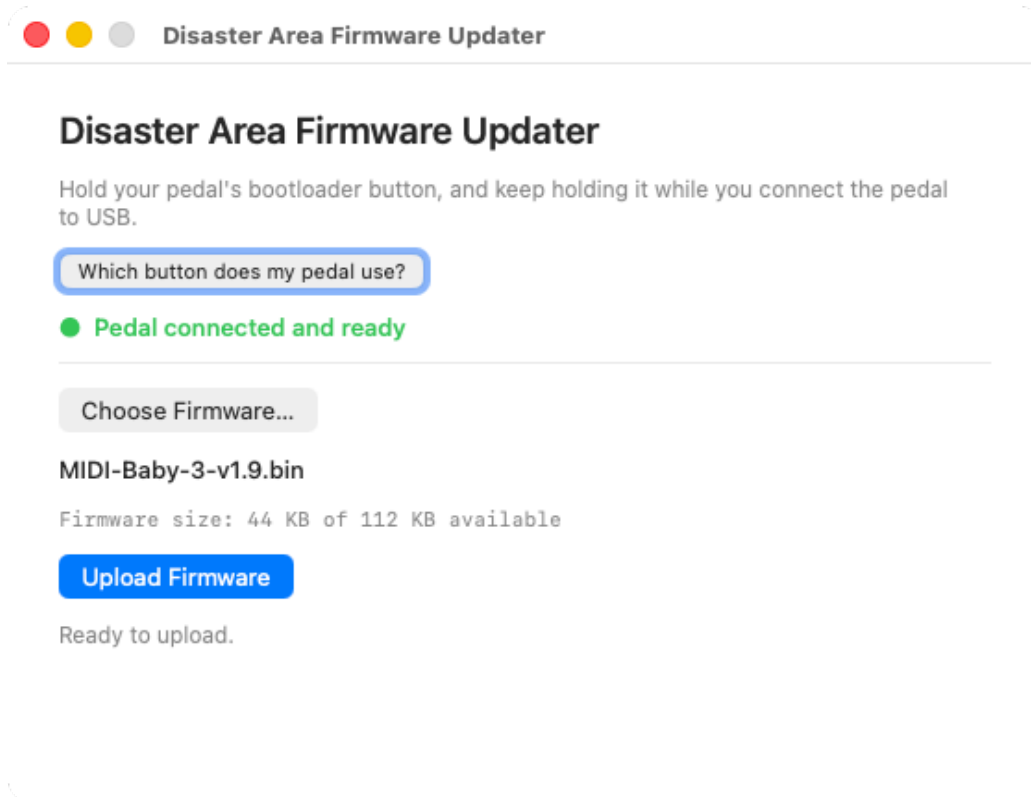


Figure 3 — The application shows the file name and the size. The size must be less than the memory that is available.

1. Click **Upload Firmware**.
2. Read the file name in the dialog again. This is your last check before the application writes to the pedal.

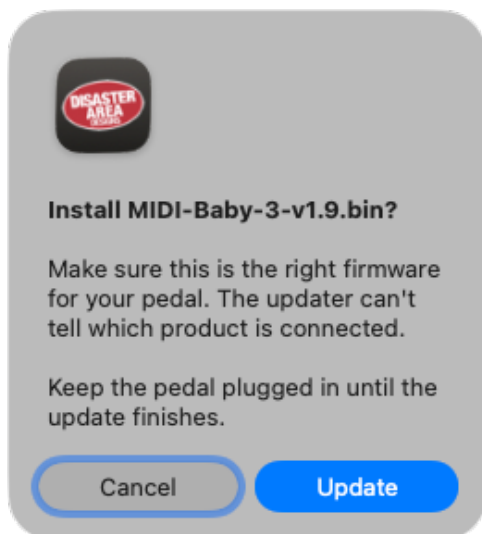


Figure 4 — The dialog shows the file name.

1. Click **Update**.
2. Wait. The bar shows the progress of the upload. The status line shows **Uploading - do not unplug the pedal....**

3. Click **OK** when the message **Upload complete** shows.

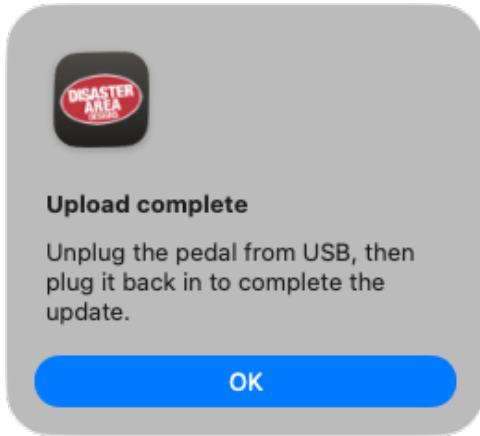


Figure 5 — The upload is complete.

1. Disconnect the pedal from USB.
2. Set the BOOT DIP switch to the RUN position. This step applies to the micro.ghost only.
3. Connect the pedal to USB again.

The pedal now operates with the new firmware.

The micro.ghost stays in bootloader mode while the switch is in the BOOT position. Therefore you must do step 14 before step 15.

The figures in this section show the Mac version. The Windows version shows the same status line, the same buttons and the same messages.

6. The first time you open the application

6.1 (Mac) Permit access to the pedal

macOS keeps removable disks safe. Thus the application must have your permission. The application asks for this permission one time.

Do these steps if the status line shows **macOS is blocking access to your pedal**:

1. Click **Open Settings** in the message.
2. Open **Privacy & Security**. Then open **Files and Folders**.
3. Find **DAD Firmware Updater** in the list.
4. Set **Removable Volumes** to on.
5. Open the application again.

6.2 (Windows) The message "Windows protected your PC"

Windows can show the message **Windows protected your PC**. This message shows because the file is new, and not because the file has a fault.

1. Click **More info**.
2. Click **Run anyway**.

Windows does not show this message again for this file.

Your antivirus program can also examine the file when you open it the first time. Let the examination become complete. Then open the application again.

7. Fault isolation

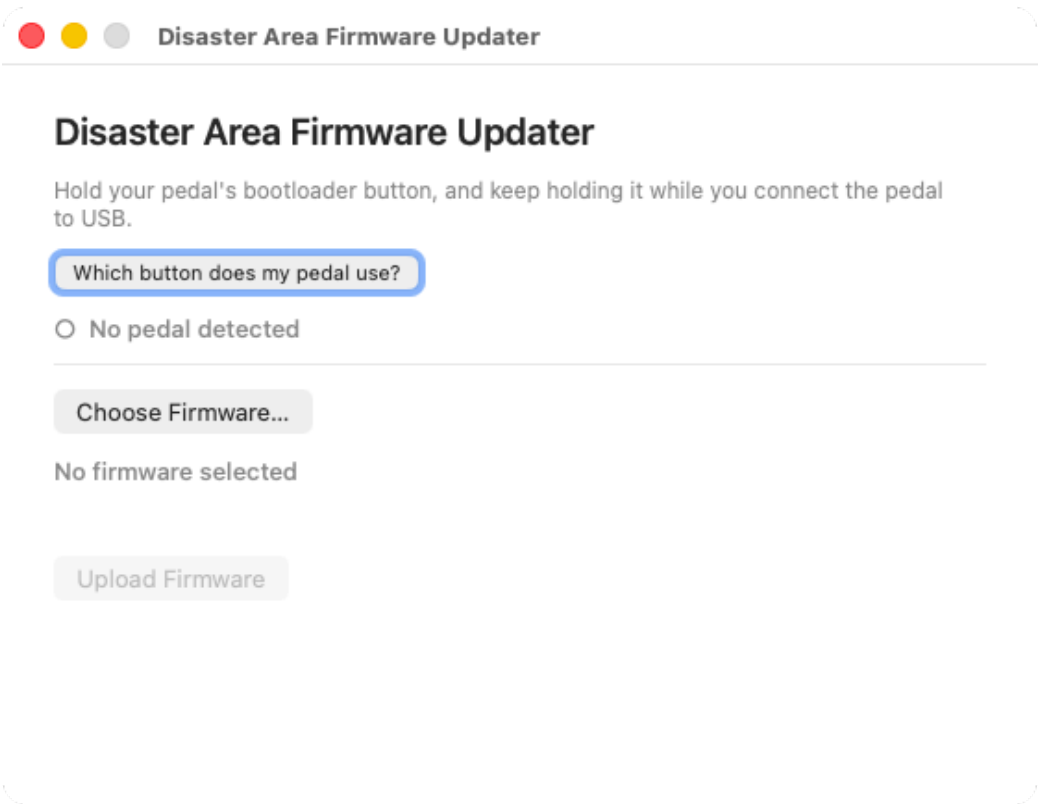


Figure 6 — The application does not find a pedal. The dot is not green. Do section 4 again.

Table 3 — Messages

Message	Version	Cause	Action
No pedal detected	Both	The pedal is not in bootloader mode. Or the cable supplies power only.	Do section 4 again. Use a cable that transmits data.
		More than	Disconnect all pedals

2 pedals connected – please connect only one	Both	one pedal is connected.	except one.
macOS is blocking access to your pedal	Mac	The application does not have permission.	Do section 6.1.
Windows is blocking access to your pedal	Windows	An antivirus program examines the drive.	Wait 10 seconds. If the message stays, disconnect the pedal and do section 4 again.
This file can't be used	Mac	The file is not firmware. Or the file is too large for the pedal.	Select the correct .bin file for your product.
This isn't a firmware file	Windows	The file is a .zip , a .hex , or a different type of file.	Open the .zip file and select the .bin file inside it.
This firmware is too big for the pedal you've connected	Windows	The firmware is for a different product.	Select the correct file for your product.
This file is too small to be firmware	Windows	The download did not become complete.	Download the firmware again.
Another program is using your pedal	Windows	A File Explorer window	Close the window. Then click

		shows the DISASTER drive.	Upload Firmware again.
Windows won't let the updater write to your pedal	Windows	An antivirus program holds the drive.	Disconnect the pedal. Wait 5 seconds. Do section 4 and section 5 again.
A warning about a different product	Both	The pedal reports a different memory size than the firmware expects.	Stop. Make sure that the firmware is for this product.
Upload failed	Both	The pedal was disconnected. Or the computer stopped the write.	Disconnect the pedal. Do section 4 and section 5 again.
The DISASTER drive stays in File Explorer after the update	Windows	The application could not eject the drive.	This is not a fault. Disconnect the pedal. The drive goes away.
The micro.ghost shows as the DISASTER disk after the update	Both	The BOOT DIP switch stays in the BOOT position.	Disconnect the pedal. Set the switch to the RUN position. Connect the pedal again.

The application examines the file before it writes. It refuses a file that is smaller than 1 KB, a file that is larger than the memory in the pedal, and a file that is not a program for this type of pedal.

If the pedal does not start after an update, do the procedure again. The bootloader stays in the pedal. Thus you can always install the firmware again.

8. Glossary

Term	Meaning
Firmware	The program that operates in the pedal.
Bootloader mode	The condition in which the pedal shows as a disk and accepts new firmware.
DISASTER disk	The disk that shows on the desktop, or in File Explorer, when the pedal is in bootloader mode.
Upload	The transmission of the firmware file from the computer to the pedal.
<code>.bin</code> file	The firmware file. This is the only type of file that the pedal accepts.